Supplementary Information:
The Lonely Runner Problem on Pair-Sum Lattices

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1 The verification record of the scaling-closure derivation

Every number above is either an exhaustive census with two independent decision procedures, or a sample with its size and interval, or a proved statement with its machine check; the records, in the order of appearance: the pool censuses (the pruned-cascade and the flattened-popcount procedures agree exactly at $p = 17/19/23$; an independent pure-enumeration brute force agrees on 21/21 sampled tuples across $p = 17/19/23/31$, covering both admissible and inadmissible branches; the $T = 9$ stride samples carry full decisions, and the $m = 6$ code path shares the cascade code brute-force-verified at four $m = 5$ primes, with the external consistency check the agreement of the projected pools with the independently measured pool scale); the zone-resolved γ table (764 pairs, five committed cells, the identical seed reproducing the committed family selections); the chain-class machine checks (123,240 solutions, 0 violations of the sum identity or the reach bound; the count reconciliations 131,520 against 123,240 and 10,080 against 8,880); the reflection fiber (survival exactly 1 at every $\bar{\lambda} = 1$ pair, every family, every cell); the renormalization-orbit law (machine-checked set-by-set at $p = 17$ and $p = 19$); the volume census and its ground truth (9/9 brute-force keys; the admissibility reproductions 840/840 and 960/1,680; the order-independence of volume across all 24 orderings); the essential-arc, budget-identity, and cap checks on the probed families; and the cascade depths at the committed cells (medians 2, maximum 4); and the character-system record of §**??** (the rigidity census over 1,678 admissible keys with every key inside the committed census scope matching it exactly, 1,654/1,654, 0 mismatches; the per-assignment renormalization bijection, 801/801 images injective with per-key counts matching under the explicit σ; the falsifications of V' and V''; the character-system aggregate invariance on both full top orbits; the chain controls 8,880/4,680/29,760 reproduced by the same code path). The transfer-matrix route's record: the generic dichotomy and trace fingerprints (1,500 systems, 12,000 comparisons, zero mismatches); the 66 homogenized pair systems on the six committed families with the closed-form orbit inventories verified by full traversal and the finite-horizon survival counts verified by direct simulation (zero mismatches); the loop-composition identity (algebraic and numeric, 500 points per family); the per-step equality of the automaton's survivor counts with the committed μ-space cascade convention (zero mismatches, all steps, all families; the committed chain-family depth reproduced); the end-to-end fiber-covering ground truth (2,800/2,800); the reflection pairs (survival fraction exactly 1, all six families, all three primes); and the reproduction of the committed volume margins (876/108/186 against $95\times$ /1,207 $\times$ /3,803$\times$). The three-rung depth record of Table **??** is calibration from the earlier cascade runs and is labeled as such wherever it is used.

2 Reproducibility and soundness record

Every empirical and classification claim in this paper rests on exact integer computation, and every computation is paired with a control or an independent reimplementation. This section records the checks; the committed scripts, logs, and experiment notes are bundled with the paper [**?**]. The record matters beyond convention here: a measurement program on 10^6-scale corpora is exactly the setting where silent bugs masquerade as findings, and the discipline below is what separates the two.

Table 1: The verification record. All counts are of exact checks with the stated outcome; "0 mismatches" means exact agreement of two independent computations.

check	scope	outcome
size law (Lem. **??**)	all residues, $N \le 111$ (T=6), $N \le 31$ (T=5)	0 violations
classification predicate $\iff$ mask covering	exhaustive over unit multisets at $7, 11, 13, 17, 19, 37$ (126–82,251 per modulus)	0 mismatches
K_{19}/K_{37} class list $\iff$ covering	all unit 4-multisets, both directions (5,985 and 82,251)	0 mismatches

Table 1 (continued)

check	scope	outcome
battery soundness (flag ⇒ certify)	exhaustive $N \le 31$ (46,371 at n=4; 324,626 at n=5); exhaustive $N \in \{37, 49, 81, 95\}$ (91,390; 270,725; 1,929,501; 3,612,280); exhaustive $N \in \{99, 111\}$ (4,249,575; 6,672,876); random 32–95 (256,000); random 96–111 (64,000)	0 violations
polygon constructive witness	sampled across all corpora	0 failures
reduced-optimum cross-check	direct grid maximization vs. bitmask, sampled	0 mismatches
lift law (Lem. **??**)	random lifts	0/2,727
equality theorem (Thm. **??**)	fresh non-circular check (folklore breakpoints vs. pair lattices), 885/885; from-scratch independent reimplementation (time domain, differences included, $n = 2$–6, $v \le 200$), with exact anchors	2,870/2,870
ground truth of (TAU-$_5$)	every primitive 5-set, $v \le 56$	3,712,576/3,712,576
trichotomy proof, line by line	every intermediate claim of Lemma T (non-wrapping, span bounds, two-lattice descent, two-block identity) at all primes $p \le 250$, plus the run lemmas at $p \le 37$	0 violations
colliding-unit ground truth	all ±-colliding $(1K, 3U)$ families, both collision topologies, $p \le 43$	3,3 M families, 0 coverings
shear linearity	per-fiber t-space offsets of colliding lifts, all v_2, m, j at $p \le 37$	0 violations of $\sigma_3 - \sigma_2 = -\bar{u}mj$
boundary finite case analyses	Lemma B at 11; Lemma A at 19: every viable configuration certified, independent brute force in agreement (and reproducing the known failures 50 at 7, 2 at 13 as witness-tuple counts; the $17, 31, 37$ analyses are redundant with the extended proof ranges)	exact
group-ring identity (Thm. **??**)	36,308 checks: all 52 Lemma A exceptions (E-profiles exactly as classified), every covering of the census ($11 \le p \le 61$) plus boundary $p = 5, 7, 13$ (12,256), 24,000 random configurations incl. non-coverings (sharp signed form; sign ⇔ covering; kernel $= \mathbb{Z}\overline{\Phi}_p$)	all pass
covering census (Prop. **??**)	all critical coverings over all $(d_2, d_3) \in (\mathbb{Z}_p^*)^2$, normal form, $p \in \{5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61\}$	four families, counts constant in p; 0 ±-distinct coverings at $p \ge 11$
foot census (Input E)	every prime $p \le 419$, both classes, every critical (s_1, s), ±-reps $d \ge 2$: one/two-foot viable sets; the M4/M5 consumed pairs; Lemma B's contained census at $(6, 6)$	consumed form clean from $p \ge 31$; now *proved* (Lemma **??**)
Lemma E machinery (Lemmas **??**–**??**)	every (p, d, s) with $p \le 500$ (45 primes, 10,410 triples): column decomposition, band and block identities, the pair-sum identity as an *exact* equality, the auxiliary bounds; the theorem direct to $p \le 2000$ (136,806) and, via the identity, to $p \le 4000$ (367,986); endpoint/dirty-prime values exact; 76/76 viable-set agreement with the foot census	all hold
parity mechanism (Prop. **??**)	every census $(2, 2)$ solution at $p \in \{23, 29, 41, 31, 37, 43\}$: integer-parity runs, non-wrapping, spills ≤ ov in A_1; per-(multiset, family) counts	100%; counts = Table **??**
pq translation (Thm. **??**)	constants; footprint formula (4,394 set equalities, 8 moduli); row census (24 resp. 312, 14 primes); $(1, 1, 2)$ cells (144,138 checks, 11 moduli 35–143, including $N = 133$); $(1K, 3U)$ cells (989,000 checks); pinning (37,672 rows, ≤ 6 good rows); T-10 reproduction at $77, 91$	all pass; 0 coverings
higher-rung machinery (Lem. **??**)	three-tier arc law vs the committed $T = 6$ tables ($p \le 37$, both classes) and at 19 primes of $T = 7/T = 8$ (all residue classes, every a_0); footprint formula vs direct Z_{p^2} enumeration; drift identity at $T \in \{7, 8\}$	957/957; 3,528/3,528
$T = 8$-critical census (Prop. **??**)	17 primes, all four odd ϵ classes, $k \le 17$: exactly 11 families, counts 288 ($\epsilon \in \{3, 7\}$) resp. 2,784 ($\epsilon \in \{1, 5\}$) constant in p; 0 ±-distinct in the mechanism range	exact

Table 1 (continued)

check	scope	outcome
$T = 7$ census (Prop. **??**)	18 primes, all six ϵ classes, $k \leq 11$: core-eleven dominance ($\approx 90\%$ at $k \geq 6$), tail decay $183 \to 43$, 0 $\pm$-distinct at $k \geq 4/k \geq 2$	exact
higher-rung cells (Prop. **??**)	$(1K, 4U)$ at $T = 7$: census-restricted ground truth (distinctness-filtered) at every prime $11 \leq p \leq 61$ — 0 coverings, best exit 2; brute-force control at $p \in \{5, 7, 11\}$ agreeing exactly at the degenerate $p = 5$ (8 coverings both ways); $(2K, 4U)$ at $T = 8$ at $17 \leq p \leq 43$ — 0 coverings; sampled controls 20K families at four primes	exact
$n = 4$ control	full record incl. the open set $= (3, 5, 8, 13)$	exact
range controls, both batteries	$V = 16/24/32/40/48$, all tiers, unflagged pair totals and decompositions	exact vs. committed logs
capability controls	zones, counts 20/68/16/64/32, safety of 48–111	exact
independent capability recounts	set-based scanner at all danger moduli (27 moduli)	exact agreement

2.1 Controls

Every experiment run begins with hard assertions against previously committed numbers before any new number is produced: the $n = 4$ corpus reproduces the committed record (including the identity of the open set); the $n = 5$ corpora at $V = 16, 24, 32, 40, 48$ reproduce every committed tier count for both batteries, the corpus-wide unflagged-pair totals and decompositions included; the capability scan reproduces the committed zones, the per-modulus covering counts, and the safety of 48–111 (the composite range extended twice, both times with the new moduli re-derived by the independent set-based scanner); and the signature-capability maps for 48–79 and 80–95 match the committed dictionaries exactly. The classifications were additionally anchored by hand: the $\mathbb{Z}_{17}$ covering of Example **??** was expanded by hand, the D_3 falsifier $(1, 1, 2, 3)$ on $\mathbb{Z}_7$ and its exclusion $N \equiv 1 \pmod 6$ were checked by hand, and the K_{19}/K_{37} covering counts were reproduced by three independent implementations (bitmask, frozenset, exponent-space).

2.2 The cross-validation discipline

The program's standing rule is that every empirical claim must be checked against committed data or a reference implementation before being cited. The rule earned its keep: across the program's lifetime it caught, before any affected number was recorded, a divisibility test written as a bitwise OR; a sampling artifact in which a 720-point time grid silently missed denominators like 11 and briefly masqueraded as counterexamples to the equality theorem; an unsound counting bound for mixed-order residues (245 flagged-but-non-certifying pairs, all caught by the flag-implies-certify sweep); an off-by-one in the capability scanner that contradicted the committed $n = 4$ zone; a flag-parsing slip; a verified-tier flag leaking into a proved tier (caught by a too-good "100%"); a diagnostic accumulating corpus-wide instead of per-tier (caught by an impossible count); a mislabeled canonicalization of cyclic translate classes (contains-0 representatives are not orbit representatives on $\mathbb{Z}_m$); and an unsound attempt to quotient the classification by exponent inversion (class reflection $e \mapsto -e$, which fixes the bad set only through $B_{-w} = B_w$ and is not a covering symmetry); an off-by-one in the minimal-arc length of the census fit test (caught by the assertion that every enumerated covering has $E \geq 0$, fixed, and then confirmed against a no-prune brute force); and a count-semantics reconciliation forced by the census run (the earlier 50/2 at $p = 7/13$ are witness-tuple counts, one per (d_2, d_3, a_2) with a completing placement, while the true solution counts are 272 normalized configurations $= 68$ set-triples and $8 = 2$ — the brute force confirms both readings, and the identity check ran on the full solution set). Each fix was verified by the control that caught it. We report these not as penance but as evidence that the numbers above survived an adversarial process.

2.3 Artifacts

The bundle [**?**] contains the equality-theorem verifications (`V1_equality_fresh_check.py` and the from-scratch `EQ_independent_verify.py`); the independent capability cross-checks (`N5_prime_scan_crosscheck.py` and the set-based scanners); the literature-search scripts and raw results; the experiment notes summarizing each stage; and the corpus and battery engines with their controls, one per stability extension, each with its full run log:

- `N5_leverage_experiment.py` (the committed $n = 5$ battery and rigid-zone scan),
- `N6_stability_experiment.py` (the $V = 32/40$ extension and the composite capability completion to $N \leq 79$),
- `N7_classify_19_37.py` (the K_{19}/K_{37} classifications and the augmented re-measurement),
- `N8_v48_extension.py` (the $V = 48$ data point, both batteries, capability to $N \leq 95$),
- `N9_v56_extension.py` (the $V = 56$ data point, both batteries, capability to $N \leq 111$; staged execution with per-stage state persistence and hard controls re-asserted at every stage),
- `ap_lemmas_verify.py` (the trichotomy/Lemma A/B verification of the companion note: exact lemma search, censuses, capacity structure, cell anchors; $p \leq 250$),
- `trichotomy_linecheck.py` (the line-by-line machine check of the trichotomy proof, including the general-arc variant),
- `pm_colliding_close.py` (the $\pm$-colliding closure: ground truth, fiber structure, shear linearity, pinning tables, pigeonhole),
- `boundary_finite_cases.py` (the boundary-prime finite case analyses with margin certificates and the independent brute-force cross-check),
- `groupring_bridge.py` (the group-ring translation verification: identity checks on all exceptions and census coverings, sharp-form sampling, kernel checks, the four-family census with per-multiset counts, defect statistics, and the witness/solution count reconciliation; output `out_groupring_bridge.json`, log `groupring_run.log`),
- `foot_census.py` (the Input E boundary census: one/two-foot viable sets at every prime $p \leq 419$, the consumed-pair cleanliness, and the Lemma B $p = 17$ form check; output `out_foot_census.json`),
- `lemmaE_master.py` (the three-distance foot-count verification, discharging Input E: column decomposition, band and block identities, the pair-sum identity as an exact equality, the auxiliary bounds of Lemma **??**, the theorem of Lemma **??** direct to $p \leq 2000$ and via the identity to $p \leq 4000$, the endpoint and dirty-prime values of Remark **??**, and the viable-set cross-check against the foot census; output `out_lemmaE.json`),
- `pq_side.py` / `pq_side2.py` / `parity_check.py` (the pq translation battery: constants and footprint formula, row censuses, $(1, 1, 2)$ and $(1K, 3U)$ cell enumeration with margins, pinning and drift analysis, and the parity-mechanism structural check; outputs `out_pq_side.json`, `out_pq_side2.json`),
- `higher_rung_step1.py` with the resumable `drive_step1.py` (the general-T machinery controls and the higher-rung censuses: the $T = 7$ four-difference census at 18 primes with per-family projections, the $T = 8$-critical census at 17 primes, the $T = 8$ five-difference probe; output `out_higher_rung_step1.json`, log `higher_rung_run1.log`),
- `higher_rung_step2.py` (the shear side at the higher rungs: the drift-law verification, the census-restricted and brute-force ground truths for the $(1K, 4U)$ cell at $T = 7$ and the $(2K, 4U)$ cell at $T = 8$, the exit-time measurements; output `out_higher_rung_step2.json`),

- `higher_rung_step3.py` (the sampled family controls for the two frontier cells; output `out_higher_rung_step3.json`),
- `pool_sat.c` with `gt_pool_brute.py` (the pool saturation censuses of §??: the m-general admissibility checker in two cross-validated implementations plus dump/resume stride sampling, and the independent pure-enumeration brute force agreeing on 21/21 sampled tuples across $p = 17/19/23/31$; dumps `dump_p{17,19,23}.tsv`, `pool_T8_p31.tsv`, `dump_T9_p{29,31}_sample.tsv`),
- `h2_pair_check.py` (the zone-resolved cut-quality measurement of Table ??: per-pair γ on the committed cells, 764 pairs; output `out_h2_pair_check.json`),
- `chain_structure_check.py` (the chain-class machine checks of Lemmas ??–?? and Theorem ??: 123,240 solutions, the sum identity, the reach bound, and the count reconciliations),
- `scaling_closure_extract.py` and `scaling_closure_verify.py` (the read-only calibration from the committed JSONs and the targeted lemma verification of §??: solution-volume probes, two-fiber dilate law, resonance zones, cascade depths at the committed cells, sign robustness; outputs `out_scaling_closure_extract.json`, `out_scaling_closure_verify.json`),
- `h1_t8_census.py`, `h1_t8_structure.py`, `h1_t8_prime.py`, `h1_t8_path.py` (the $T = 8$ volume census, mechanism probes, and prime extensions of §??, with the independent brute-force ground truth (9/9 keys) and the renormalization-orbit verification; states `h1_t8_state.json` / `h1_t8_prime_state.json`, outputs `out_h1_t8_census.json`, `out_h1_t8_structure.json`, `out_h1_t8_prime.json`).

All runs are deterministic; every number in §??–2 appears verbatim in a committed log.

2.4 Data and code availability

The complete artifact bundle (scripts, committed logs, JSON records, and the figures) accompanies this paper and is deposited for public access together with the PDF; the sources build with `tectonic` and a single command. The verification scripts have no dependencies beyond CPython 3.12 with `numpy`; each re-derives its controls from first principles and asserts them before producing any new number, so any single experiment can be re-verified in isolation. This document and the bundle are released under the Creative Commons Attribution 4.0 license (CC BY 4.0), the better to allow the community to re-run, refute, or extend the record.